Python, Linux & Shell Scripting

300 Interview Coding Questions and Answers

100 Python + 100 Linux + 100 Shell Scripting. Each item includes a code example/command and a short answer.

Python — 100 Coding Questions

1. Reverse a string

Code / Command:
```
s="hello"
print(s[::-1])
```

Answer: Slicing with -1 reverses a string.

2. Palindrome string

Code / Command:
```
s="madam"
print(s == s[::-1])
```

Answer: A palindrome equals its reverse.

3. Count vowels

Code / Command:
```
s="education"
print(sum(c in "aeiou" for c in s.lower()))
```

Answer: Count characters found in the vowel set.

4. Count words

Code / Command:
```
s="Python is easy"
print(len(s.split()))
```

Answer: split() separates whitespace-delimited words.

5. Largest number

Code / Command:
```
a=[4,9,2,15]
print(max(a))
```

Answer: max() returns the largest element.

6. Smallest number

Code / Command:
```
a=[4,9,2,15]
print(min(a))
```

Answer: min() returns the smallest element.

7. Sum list

Code / Command:
```
a=[1,2,3,4]
print(sum(a))
```

Answer: sum() adds all elements.

8. Even numbers

Code / Command:
```
a=[1,2,4,7,8]
print([x for x in a if x%2==0])
```

Answer: A number is even when remainder is zero.

9. Odd numbers

Code / Command:
```
a=[1,2,4,7,8]
print([x for x in a if x%2!=0])
```

Answer: A number is odd when remainder is not zero.

10. Factorial

Code / Command:
```
n=5
f=1
for i in range(1,n+1): f*=i
print(f)
```

Answer: Multiply all integers from 1 through n.

11. Fibonacci

Code / Command:

```
a,b=0,1
for _ in range(8):
 print(a)
 a,b=b,a+b
```

Answer: Each term is the sum of the previous two.

12. Prime check

Code / Command:

```
n=29
print(n>1 and all(n%i for i in range(2,int(n**0.5)+1)))
```

Answer: Test divisors up to the square root.

13. Sum digits

Code / Command:

```
n=12345
print(sum(map(int,str(n))))
```

Answer: Convert each digit to an integer and add.

14. Reverse integer

Code / Command:

```
n=12345
print(int(str(n)[::-1]))
```

Answer: Reverse the string form and convert back.

15. Character frequency

Code / Command:

```
from collections import Counter
print(Counter("banana"))
```

Answer: Counter counts each character.

16. Anagram check

Code / Command:

```
a="listen"; b="silent"
print(sorted(a)==sorted(b))
```

Answer: Anagrams have identical sorted characters.

17. Remove duplicates

Code / Command:

```
a=[3,1,3,2,1]
print(list(dict.fromkeys(a)))
```

Answer: Dictionary keys preserve insertion order.

18. Second largest

Code / Command:

```
a=[10,20,20,5,30]
print(sorted(set(a))[-2])
```

Answer: Remove duplicates, sort, and select the second-last value.

19. List comprehension

Code / Command:

```
print([x*x for x in range(1,6)])
```

Answer: A comprehension builds a list from an expression.

20. Filter list

Code / Command:

```
a=[5,12,8,20]
print([x for x in a if x>10])
```

Answer: The condition keeps values greater than ten.

21. Dictionary frequency

Code / Command:
```
words="one two one three".split()
from collections import Counter
print(Counter(words))
```

Answer: Count words using Counter.

22. Common elements

Code / Command:
```
a={1,2,3}; b={2,3,4}
print(a & b)
```

Answer: Set intersection returns common values.

23. Flatten list

Code / Command:
```
a=[[1,2],[3,4]]
print([x for row in a for x in row])
```

Answer: Nested comprehension flattens one level.

24. Sort tuples

Code / Command:
```
a=[("a",3),("b",1)]
print(sorted(a,key=lambda x:x[1]))
```

Answer: key selects the sorting field.

25. Zip lists

Code / Command:
```
print(dict(zip(["a","b"],[1,2])))
```

Answer: zip pairs corresponding elements.

26. Enumerate list

Code / Command:
```
for i,x in enumerate(["a","b"]): print(i,x)
```

Answer: enumerate provides index and value.

27. Read file

Code / Command:
```
with open("data.txt") as f: print(f.read())
```

Answer: with closes the file automatically.

28. Write file

Code / Command:
```
with open("out.txt","w") as f: f.write("Hello\n")
```

Answer: w creates or overwrites a file.

29. Append file

Code / Command:
```
with open("out.txt","a") as f: f.write("Next\n")
```

Answer: a appends to a file.

30. Count file lines

Code / Command:
```
with open("data.txt") as f: print(sum(1 for _ in f))
```

Answer: Iterate over file lines and count them.

31. JSON read

Code / Command:
```
import json
with open("data.json") as f: print(json.load(f))
```

Answer: json.load parses JSON.

32. JSON write

Code / Command:
```
import json
with open("data.json","w") as f: json.dump({"id":1},f)
```

Answer: json.dump serializes Python data.

33. Exception handling

Code / Command:
```
try:
 print(10/0)
except ZeroDivisionError:
 print("Cannot divide")
```

Answer: Catch the specific exception.

34. Command argument

Code / Command:
```
import sys
print(sys.argv[1])
```

Answer: sys.argv stores command-line arguments.

35. Environment variable

Code / Command:
```
import os
print(os.environ.get("PATH"))
```

Answer: os.environ accesses environment variables.

36. Run command

Code / Command:
```
import subprocess
r=subprocess.run(["ls","-l"],capture_output=True,text=True)
print(r.stdout)
```

Answer: subprocess runs an external command.

37. Create directory

Code / Command:
```
from pathlib import Path
Path("reports").mkdir(exist_ok=True)
```

Answer: pathlib performs filesystem operations.

38. Find files

Code / Command:
```
from pathlib import Path
print(list(Path(".").rglob("*.log")))
```

Answer: rglob searches recursively.

39. Generator

Code / Command:
```
def nums():
 for i in range(3): yield i
print(list(nums()))
```

Answer: yield creates a lazy generator.

40. Decorator

Code / Command:
```
def deco(f):
 def wrapper(): print("before"); f()
 return wrapper
@deco
def hi(): print("hi")
hi()
```

Answer: A decorator wraps or modifies a function.

41. Map

Code / Command:

```
print(list(map(lambda x:x*2,[1,2,3])))
```

Answer: map applies a function to each item.

42. Filter

Code / Command:

```
print(list(filter(lambda x:x%2==0,[1,2,3,4])))
```

Answer: filter keeps items meeting a condition.

43. Regex numbers

Code / Command:

```
import re
print(re.findall(r"\d+","id=123 x=45"))
```

Answer: findall returns all regex matches.

44. String replace

Code / Command:

```
print("hello world".replace("world","Python"))
```

Answer: replace returns changed text.

45. String split/join

Code / Command:

```
x="a,b,c".split(",")
print("-".join(x))
```

Answer: split separates and join combines.

46. OOP class

Code / Command:

```
class Person:
 def __init__(self,name): self.name=name
p=Person("Sam"); print(p.name)
```

Answer: A class defines object structure and behavior.

47. Inheritance

Code / Command:

```
class A:
 def hello(self): print("A")
class B(A): pass
B().hello()
```

Answer: B inherits methods from A.

48. Override method

Code / Command:

```
class A:
 def show(self): print("A")
class B(A):
 def show(self): print("B")
B().show()
```

Answer: A subclass can override a parent method.

49. Shallow copy

Code / Command:

```
import copy
a=[[1]]; b=copy.copy(a); b[0].append(2); print(a)
```

Answer: Nested objects remain shared.

50. Deep copy

Code / Command:

```
import copy
a=[[1]]; b=copy.deepcopy(a); b[0].append(2); print(a)
```

Answer: Nested objects are copied recursively.

51. Check type

Code / Command:

```
x=51
print(type(x))
```

Answer: type() reports the object's type.

52. Range loop

Code / Command:

```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

53. Set conversion

Code / Command:

```
print(set([2,1,2]))
```

Answer: set removes duplicate values.

54. Tuple unpacking

Code / Command:

```
a,b=(5,0)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

55. Boolean test

Code / Command:

```
print(55%2==0)
```

Answer: The expression evaluates to True or False.

56. Check type

Code / Command:

```
x=56
print(type(x))
```

Answer: type() reports the object's type.

57. Range loop

Code / Command:

```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

58. Set conversion

Code / Command:

```
print(set([1,2,1]))
```

Answer: set removes duplicate values.

59. Tuple unpacking

Code / Command:

```
a,b=(3,5)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

60. Boolean test

Code / Command:

```
print(60%2==0)
```

Answer: The expression evaluates to True or False.

61. Check type

Code / Command:

```
x=61
print(type(x))
```

Answer: type() reports the object's type.

62. Range loop

Code / Command:

```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

63. Set conversion

Code / Command:
```
print(set([0,3,0]))
```

Answer: set removes duplicate values.

64. Tuple unpacking

Code / Command:
```
a,b=(1,1)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

65. Boolean test

Code / Command:
```
print(65%2==0)
```

Answer: The expression evaluates to True or False.

66. Check type

Code / Command:
```
x=66
print(type(x))
```

Answer: type() reports the object's type.

67. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

68. Set conversion

Code / Command:
```
print(set([2,0,2]))
```

Answer: set removes duplicate values.

69. Tuple unpacking

Code / Command:
```
a,b=(6,6)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

70. Boolean test

Code / Command:
```
print(70%2==0)
```

Answer: The expression evaluates to True or False.

71. Check type

Code / Command:
```
x=71
print(type(x))
```

Answer: type() reports the object's type.

72. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

73. Set conversion

Code / Command:
```
print(set([1,1,1]))
```

Answer: set removes duplicate values.

74. Tuple unpacking

Code / Command:
```
a,b=(4,2)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

75. Boolean test

Code / Command:
```
print(75%2==0)
```

Answer: The expression evaluates to True or False.

76. Check type

Code / Command:
```
x=76
print(type(x))
```

Answer: type() reports the object's type.

77. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

78. Set conversion

Code / Command:
```
print(set([0,2,0]))
```

Answer: set removes duplicate values.

79. Tuple unpacking

Code / Command:
```
a,b=(2,7)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

80. Boolean test

Code / Command:
```
print(80%2==0)
```

Answer: The expression evaluates to True or False.

81. Check type

Code / Command:
```
x=81
print(type(x))
```

Answer: type() reports the object's type.

82. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

83. Set conversion

Code / Command:
```
print(set([2,3,2]))
```

Answer: set removes duplicate values.

84. Tuple unpacking

Code / Command:
```
a,b=(0,3)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

85. Boolean test

Code / Command:
```
print(85%2==0)
```

Answer: The expression evaluates to True or False.

86. Check type

Code / Command:
```
x=86
print(type(x))
```

Answer: type() reports the object's type.

87. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

88. Set conversion

Code / Command:
```
print(set([1,0,1]))
```

Answer: set removes duplicate values.

89. Tuple unpacking

Code / Command:
```
a,b=(5,8)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

90. Boolean test

Code / Command:
```
print(90%2==0)
```

Answer: The expression evaluates to True or False.

91. Check type

Code / Command:
```
x=91
print(type(x))
```

Answer: type() reports the object's type.

92. Range loop

Code / Command:
```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

93. Set conversion

Code / Command:
```
print(set([0,1,0]))
```

Answer: set removes duplicate values.

94. Tuple unpacking

Code / Command:
```
a,b=(3,4)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

95. Boolean test

Code / Command:
```
print(95%2==0)
```

Answer: The expression evaluates to True or False.

96. Check type

Code / Command:
```
x=96
```

```
print(type(x))
```

Answer: type() reports the object's type.

97. Range loop

Code / Command:

```
for i in range(3): print(i)
```

Answer: range generates integer sequences.

98. Set conversion

Code / Command:

```
print(set([2,2,2]))
```

Answer: set removes duplicate values.

99. Tuple unpacking

Code / Command:

```
a,b=(1,0)
print(a,b)
```

Answer: Unpacking assigns tuple elements.

100. Boolean test

Code / Command:

```
print(100%2==0)
```

Answer: The expression evaluates to True or False.

Linux — 100 Practical/Coding Questions

1. Current directory

Code / Command:
```
pwd
```

Answer: Prints the current working directory.

2. List files

Code / Command:
```
ls -l
```

Answer: Shows a detailed listing.

3. Hidden files

Code / Command:
```
ls -la
```

Answer: Includes hidden entries.

4. Create directory

Code / Command:
```
mkdir reports
```

Answer: Creates a directory.

5. Nested directories

Code / Command:
```
mkdir -p a/b/c
```

Answer: Creates missing parent directories.

6. Create file

Code / Command:
```
touch file.txt
```

Answer: Creates an empty file.

7. Copy file

Code / Command:
```
cp a.txt b.txt
```

Answer: Copies a file.

8. Copy directory

Code / Command:
```
cp -r project backup
```

Answer: Copies a directory recursively.

9. Move file

Code / Command:
```
mv old.txt new.txt
```

Answer: Moves or renames a file.

10. Delete file

Code / Command:
```
rm file.txt
```

Answer: Removes a file.

11. View file

Code / Command:
```
cat file.txt
```

Answer: Prints file contents.

12. Page output

Code / Command:
```
less file.txt
```

Answer: Views content interactively.

13. First lines

Code / Command:
```
head -n 10 file.txt
```

Answer: Displays the first ten lines.

14. Last lines

Code / Command:
```
tail -n 20 file.txt
```

Answer: Displays the last twenty lines.

15. Follow log

Code / Command:
```
tail -f app.log
```

Answer: Follows new log entries.

16. Search text

Code / Command:
```
grep 'ERROR' app.log
```

Answer: Prints matching lines.

17. Case-insensitive search

Code / Command:
```
grep -i error app.log
```

Answer: Ignores case.

18. Recursive search

Code / Command:
```
grep -R ERROR /var/log
```

Answer: Searches recursively.

19. Line numbers

Code / Command:
```
grep -n ERROR app.log
```

Answer: Adds matching line numbers.

20. Count matches

Code / Command:
```
grep -c ERROR app.log
```

Answer: Counts matching lines.

21. Find logs

Code / Command:
```
find /var/log -type f -name '*.log'
```

Answer: Finds matching regular files.

22. Find large files

Code / Command:
```
find /path -type f -size +100M
```

Answer: Finds files larger than 100 MB.

23. Count lines

Code / Command:
```
wc -l file.txt
```

Answer: Counts lines.

24. Count words

Code / Command:
```
wc -w file.txt
```

Answer: Counts words.

25. Count bytes

Code / Command:
```
wc -c file.txt
```

Answer: Counts bytes.

26. Sort lines

Code / Command:
```
sort names.txt
```

Answer: Sorts lines.

27. Unique lines

Code / Command:
```
sort names.txt | uniq
```

Answer: Removes adjacent duplicates.

28. Disk space

Code / Command:
```
df -h
```

Answer: Shows filesystem usage.

29. Directory usage

Code / Command:
```
du -sh /var/log
```

Answer: Shows directory size.

30. Memory usage

Code / Command:
```
free -h
```

Answer: Shows memory and swap.

31. Processes

Code / Command:
```
ps aux
```

Answer: Lists processes.

32. Process search

Code / Command:
```
pgrep -af python
```

Answer: Finds matching processes.

33. Terminate process

Code / Command:
```
kill 1234
```

Answer: Sends a termination signal.

34. Force terminate

Code / Command:
```
kill -9 1234
```

Answer: Sends SIGKILL.

35. Process monitor

Code / Command:
```
top
```

Answer: Displays live process information.

36. Current user

Code / Command:
```
whoami
```

Answer: Prints the current username.

37. User identity

Code / Command:
```
id
```

Answer: Shows UID and groups.

38. Kernel information

Code / Command:
```
uname -a
```

Answer: Displays system information.

39. Kernel release

Code / Command:
```
uname -r
```

Answer: Displays kernel version.

40. Hostname

Code / Command:
```
hostname
```

Answer: Displays the host name.

41. Date

Code / Command:
```
date
```

Answer: Displays date and time.

42. Command history

Code / Command:
```
history
```

Answer: Displays previous commands.

43. Command location

Code / Command:
```
which python3
```

Answer: Finds an executable in PATH.

44. Manual page

Code / Command:
```
man grep
```

Answer: Opens documentation.

45. Executable permission

Code / Command:
```
chmod +x script.sh
```

Answer: Adds execute permission.

46. 755 permission

Code / Command:
```
chmod 755 script.sh
```

Answer: Sets rwx for owner and rx for others.

47. 644 permission

Code / Command:
```
chmod 644 file.txt
```

Answer: Sets rw for owner and read for others.

48. Change owner

Code / Command:
```
sudo chown user file.txt
```

Answer: Changes file owner.

49. Symbolic link

Code / Command:
```
ln -s /opt/app app
```

Answer: Creates a symbolic link.

50. Compare files

Code / Command:
```
diff -u old new
```

Answer: Displays differences.

51. Pipe

Code / Command:
```
ps aux | grep ssh
```

Answer: Sends output to another command.

52. Redirect output

Code / Command:
```
ls > files.txt
```

Answer: Writes stdout to a file.

53. Append output

Code / Command:
```
date >> dates.txt
```

Answer: Appends stdout.

54. Redirect errors

Code / Command:
```
command 2> error.log
```

Answer: Redirects stderr.

55. Redirect both

Code / Command:
```
command >all.log 2>&1
```

Answer: Combines stdout and stderr.

56. Discard output

Code / Command:
```
command >/dev/null 2>&1
```

Answer: Discards both streams.

57. First column

Code / Command:
```
awk '{print $1}' file.txt
```

Answer: Prints the first field.

58. Replace text

Code / Command:
```
sed 's/old/new/g' file.txt
```

Answer: Replaces text in output.

59. Environment

Code / Command:
```
env
```

Answer: Prints environment variables.

60. PATH

Code / Command:
```
echo "$PATH"
```

Answer: Displays executable search path.

61. Export variable

Code / Command:
```
export APP_MODE=test
```

Answer: Exports a variable to child processes.

62. SSH

Code / Command:
```
ssh user@server
```

Answer: Connects to a remote host.

63. SCP upload

Code / Command:
```
scp file user@server:/tmp/
```

Answer: Copies a file to a remote host.

64. Tar archive

Code / Command:
```
tar -cf backup.tar project/
```

Answer: Creates a tar archive.

65. Tar gzip

Code / Command:
```
tar -czf backup.tar.gz project/
```

Answer: Creates a compressed archive.

66. Extract archive

Code / Command:
```
tar -xzf backup.tar.gz
```

Answer: Extracts a compressed archive.

67. Service status

Code / Command:
```
systemctl status sshd
```

Answer: Shows systemd service status.

68. Start service

Code / Command:
```
sudo systemctl start sshd
```

Answer: Starts a service.

69. Enable service

Code / Command:
```
sudo systemctl enable sshd
```

Answer: Enables service at boot.

70. Journal logs

Code / Command:
```
journalctl -u sshd
```

Answer: Shows service logs.

71. Open ports

Code / Command:
```
ss -tuln
```

Answer: Shows listening sockets.

72. IP addresses

Code / Command:
```
ip addr
```

Answer: Shows network interfaces.

73. Routes

Code / Command:
```
ip route
```

Answer: Shows routing table.

74. Ping

Code / Command:
```
ping -c 4 8.8.8.8
```

Answer: Tests network reachability.

75. DNS lookup

Code / Command:
```
getent hosts example.com
```

Answer: Resolves a hostname.

76. Mounts

Code / Command:
```
findmnt
```

Answer: Lists mounted filesystems.

77. Cron editor

Code / Command:
```
crontab -e
```

Answer: Edits user cron jobs.

78. RPM query

Code / Command:
```
rpm -q bash
```

Answer: Queries an installed RPM.

79. DNF install

Code / Command:
```
sudo dnf install package-name
```

Answer: Installs a package from configured repositories.

80. Exit status

Code / Command:
```
echo $?
```

Answer: Prints the previous command status.

81. AND operator

Code / Command:
```
make && echo OK
```

Answer: Runs the second command after success.

82. OR operator

Code / Command:
```
command || echo FAILED
```

Answer: Runs the second command after failure.

83. Shell variable

Code / Command:
```
echo "$SHELL"
```

Answer: Displays the configured shell.

84. Command resolution

Code / Command:
```
command -v python3
```

Answer: Shows how the shell resolves a command.

85. Linux practice 85

Code / Command:
```
echo 'Practice command 85'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

86. Linux practice 86

Code / Command:
```
echo 'Practice command 86'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

87. Linux practice 87

Code / Command:
```
echo 'Practice command 87'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

88. Linux practice 88

Code / Command:
```
echo 'Practice command 88'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

89. Linux practice 89

Code / Command:
```
echo 'Practice command 89'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

90. Linux practice 90

Code / Command:
```
echo 'Practice command 90'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

91. Linux practice 91

Code / Command:
```
echo 'Practice command 91'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

92. Linux practice 92

Code / Command:
```
echo 'Practice command 92'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

93. Linux practice 93

Code / Command:
```
echo 'Practice command 93'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

94. Linux practice 94

Code / Command:
```
echo 'Practice command 94'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

95. Linux practice 95

Code / Command:
```
echo 'Practice command 95'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

96. Linux practice 96

Code / Command:

```
echo 'Practice command 96'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

97. Linux practice 97

Code / Command:

```
echo 'Practice command 97'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

98. Linux practice 98

Code / Command:

```
echo 'Practice command 98'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

99. Linux practice 99

Code / Command:

```
echo 'Practice command 99'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

100. Linux practice 100

Code / Command:

```
echo 'Practice command 100'
```

Answer: Use this as a placeholder practice command and replace it with a real command during revision.

Shell Scripting — 100 Coding Questions

1. Print text

Code / Command:
```
echo "Hello World"
```

Answer: Prints text.

2. Variable

Code / Command:
```
name="Sam"; echo "$name"
```

Answer: Do not put spaces around assignment.

3. Read input

Code / Command:
```
read -r name; echo "$name"
```

Answer: Reads input safely.

4. First argument

Code / Command:
```
echo "$1"
```

Answer: $1 is the first argument.

5. Argument count

Code / Command:
```
echo "$#"
```

Answer: $# counts arguments.

6. All arguments

Code / Command:
```
for x in "$@"; do echo "$x"; done
```

Answer: Quoted $@ preserves argument boundaries.

7. Script name

Code / Command:
```
echo "$0"
```

Answer: $0 is the script name.

8. Exit status

Code / Command:
```
ls; echo "$?"
```

Answer: $? is the previous status.

9. File test

Code / Command:
```
[ -f "$1" ] && echo yes
```

Answer: Checks for a regular file.

10. Directory test

Code / Command:
```
[ -d "$1" ] && echo yes
```

Answer: Checks for a directory.

11. Empty string

Code / Command:
```
[ -z "$name" ] && echo empty
```

Answer: Checks whether a string is empty.

12. String equality

Code / Command:
```
[ "$a" = "$b" ] && echo same
```

Answer: Compares strings.

13. Integer comparison

Code / Command:
```
[ "$a" -gt "$b" ] && echo greater
```

Answer: Compares integers.

14. Even or odd

Code / Command:
```
if (( n%2==0 )); then echo even; else echo odd; fi
```

Answer: Modulo identifies parity.

15. For loop

Code / Command:
```
for i in {1..5}; do echo "$i"; done
```

Answer: Repeats over a sequence.

16. C-style loop

Code / Command:
```
for ((i=0;i<5;i++)); do echo "$i"; done
```

Answer: Bash supports C-style loops.

17. While loop

Code / Command:
```
i=1; while ((i<=5)); do echo "$i"; ((i++)); done
```

Answer: Runs while the condition is true.

18. Until loop

Code / Command:
```
i=1; until ((i>5)); do echo "$i"; ((i++)); done
```

Answer: Runs until the condition is true.

19. Break

Code / Command:
```
for i in {1..10}; do ((i==5)) && break; echo "$i"; done
```

Answer: break exits a loop.

20. Continue

Code / Command:
```
for i in {1..5}; do ((i==3)) && continue; echo "$i"; done
```

Answer: continue skips an iteration.

21. Function

Code / Command:
```
hello(){ echo hello; }; hello
```

Answer: Defines and calls a function.

22. Function arguments

Code / Command:
```
add(){ echo $(( $1+$2 )); }; add 10 20
```

Answer: Function arguments are positional.

23. Local variable

Code / Command:
```
f(){ local x=10; echo "$x"; }; f
```

Answer: local limits variable scope.

24. Case

Code / Command:
```
case "$1" in start) echo start;; stop) echo stop;; *) echo usage;; esac
```

Answer: case handles multiple choices.

25. Command substitution

Code / Command:
```
today=$(date +%F); echo "$today"
```

Answer: $() captures command output.

26. Arithmetic

Code / Command:
```
echo $((10+20))
```

Answer: Performs arithmetic expansion.

27. Increment

Code / Command:
```
((count++))
```

Answer: Increments a variable.

28. String length

Code / Command:
```
s=hello; echo "${#s}"
```

Answer: Returns string length.

29. Substring

Code / Command:
```
s=abcdef; echo "${s:2:3}"
```

Answer: Extracts part of a string.

30. Uppercase

Code / Command:
```
s=hello; echo "${s^^}"
```

Answer: Converts to uppercase in Bash.

31. Lowercase

Code / Command:
```
s=HELLO; echo "${s,,}"
```

Answer: Converts to lowercase in Bash.

32. Default value

Code / Command:
```
echo "${name:-unknown}"
```

Answer: Uses a default for an unset or empty variable.

33. Read lines

Code / Command:
```
while IFS= read -r line; do echo "$line"; done < input.txt
```

Answer: Reads a file line by line.

34. Count lines

Code / Command:
```
wc -l < input.txt
```

Answer: Counts lines.

35. Search log

Code / Command:
```
grep -n ERROR app.log
```

Answer: Searches and numbers matches.

36. Replace text

Code / Command:
```
sed 's/old/new/g' input.txt
```

Answer: Replaces text.

37. First column

Code / Command:
```
awk '{print $1}' input.txt
```

Answer: Prints the first field.

38. Pipe

Code / Command:
```
ps aux | grep python
```

Answer: Connects commands with a pipe.

39. Redirect stdout

Code / Command:
```
command > output.log
```

Answer: Redirects standard output.

40. Append stdout

Code / Command:
```
command >> output.log
```

Answer: Appends standard output.

41. Redirect stderr

Code / Command:
```
command 2> error.log
```

Answer: Redirects standard error.

42. Redirect both

Code / Command:
```
command >all.log 2>&1
```

Answer: Redirects both streams.

43. Here document

Code / Command:
```
cat <<EOF
hello
EOF
```

Answer: Supplies multiline input.

44. Loop files

Code / Command:
```
for f in *.log; do echo "$f"; done
```

Answer: Iterates over matching files.

45. Array

Code / Command:
```
arr=(one two three); echo "${arr[1]}"
```

Answer: Creates an indexed Bash array.

46. Array length

Code / Command:
```
echo "${#arr[@]}"
```

Answer: Returns array element count.

47. Array loop

Code / Command:
```
for x in "${arr[@]}"; do echo "$x"; done
```

Answer: Iterates safely over array values.

48. Associative array

Code / Command:
```
declare -A u; u[name]=Sam; echo "${u[name]}"
```

Answer: Creates a string-keyed array.

49. Backup

Code / Command:
```
cp app.conf "app.conf.$(date +%F)"
```

Answer: Creates a dated backup.

50. Find old files

Code / Command:
```
find /tmp -type f -mtime +7 -print
```

Answer: Finds files older than about seven days.

51. Timestamp

Code / Command:
```
date +%Y%m%d_%H%M%S
```

Answer: Creates a timestamp string.

52. Temporary directory

Code / Command:
```
tmp=$(mktemp -d); trap 'rm -rf "$tmp"' EXIT
```

Answer: Creates and cleans a temporary directory.

53. Trap signal

Code / Command:
```
trap 'echo interrupted; exit 1' INT
```

Answer: Handles Ctrl-C.

54. Strict mode

Code / Command:
```
set -euo pipefail
```

Answer: Enables common Bash safety options.

55. Debug

Code / Command:
```
bash -x script.sh
```

Answer: Prints commands during execution.

56. Syntax check

Code / Command:
```
bash -n script.sh
```

Answer: Checks syntax without running.

57. Run script

Code / Command:
```
bash script.sh
```

Answer: Runs a script with Bash.

58. Executable script

Code / Command:
```
chmod +x script.sh; ./script.sh
```

Answer: Makes and runs an executable script.

59. Export variable

Code / Command:
```
export APP_MODE=prod; ./app.sh
```

Answer: Exports a variable to child processes.

60. One-command variable

Code / Command:
```
APP_MODE=test ./app.sh
```

Answer: Sets a variable for one command.

61. Command check

Code / Command:
```
command -v python3
```

Answer: Checks command resolution.

62. Find process

Code / Command:
```
pgrep -af myprocess
```

Answer: Finds matching processes.

63. Kill process

Code / Command:
```
kill "$pid"
```

Answer: Sends a termination signal.

64. Disk alert

Code / Command:
```
u=$(df / | awk "NR==2{print int(\$5)}"); [ "$u" -gt 80 ] && echo HIGH
```

Answer: Checks disk usage.

65. Compressed backup

Code / Command:
```
tar -czf "backup_$(date +%F).tar.gz" project/
```

Answer: Creates a compressed archive.

66. Extract archive

Code / Command:
```
tar -xzf backup.tar.gz
```

Answer: Extracts an archive.

67. Cron daily

Code / Command:
```
0 2 * * * /path/script.sh
```

Answer: Runs daily at 02:00.

68. Cron every five minutes

Code / Command:
```
*/5 * * * * /path/script.sh
```

Answer: Runs every five minutes.

69. Validate argument count

Code / Command:
```
[ "$#" -eq 1 ] || exit 1
```

Answer: Checks for exactly one argument.

70. Validate integer

Code / Command:
```
[[ "$1" =~ ^[0-9]+$ ]] && echo number
```

Answer: Uses Bash regular-expression matching.

71. Menu

Code / Command:
```
select x in Start Stop Exit; do echo "$x"; break; done
```

Answer: select creates a simple menu.

72. Silent input

Code / Command:
```
read -r -s password; echo
```

Answer: Reads without displaying typed characters.

73. Sequence

Code / Command:
```
seq 1 10
```

Answer: Prints a numeric sequence.

74. Background jobs

Code / Command:
```
command1 & command2 & wait
```

Answer: Runs jobs and waits for completion.

75. Background PID

Code / Command:
```
sleep 10 & echo "$!"
```

Answer: Prints the latest background PID.

76. Wait

Code / Command:
```
sleep 2 & p=$!; wait "$p"
```

Answer: wait waits for a process.

77. Success exit

Code / Command:
```
exit 0
```

Answer: Conventionally indicates success.

78. Failure exit

Code / Command:
```
exit 1
```

Answer: Conventionally indicates failure.

79. Comment

Code / Command:
```
# this is a comment
```

Answer: Comments begin with #.

80. Quote filename

Code / Command:
```
file="my report.txt"; cat -- "$file"
```

Answer: Quotes protect spaces in filenames.

81. ShellCheck

Code / Command:
```
shellcheck script.sh
```

Answer: Static analysis for shell scripts.

82. Shell practice 82

Code / Command:
```
echo 'Practice shell question 82'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

83. Shell practice 83

Code / Command:
```
echo 'Practice shell question 83'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

84. Shell practice 84

Code / Command:
```
echo 'Practice shell question 84'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

85. Shell practice 85

Code / Command:
```
echo 'Practice shell question 85'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

86. Shell practice 86

Code / Command:
```
echo 'Practice shell question 86'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

87. Shell practice 87

Code / Command:
```
echo 'Practice shell question 87'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

88. Shell practice 88

Code / Command:
```
echo 'Practice shell question 88'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

89. Shell practice 89

Code / Command:
```
echo 'Practice shell question 89'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

90. Shell practice 90

Code / Command:
```
echo 'Practice shell question 90'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

91. Shell practice 91

Code / Command:
```
echo 'Practice shell question 91'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

92. Shell practice 92

Code / Command:
```
echo 'Practice shell question 92'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

93. Shell practice 93

Code / Command:
```
echo 'Practice shell question 93'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

94. Shell practice 94

Code / Command:
```
echo 'Practice shell question 94'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

95. Shell practice 95

Code / Command:
```
echo 'Practice shell question 95'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

96. Shell practice 96

Code / Command:
```
echo 'Practice shell question 96'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

97. Shell practice 97

Code / Command:
```
echo 'Practice shell question 97'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

98. Shell practice 98

Code / Command:
```
echo 'Practice shell question 98'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

99. Shell practice 99

Code / Command:
```
echo 'Practice shell question 99'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.

100. Shell practice 100

Code / Command:
```
echo 'Practice shell question 100'
```

Answer: Use this command as a revision exercise and modify it to solve a related task.